

Ghz wireless LAN signals. Software-defined radios, by contrast, can tune dynamically over a wider range of frequencies. A software-defined radio can receive or transmit different kinds of wireless transmissions automatically. If it is a so-called “agile radio,” it could adapt to the local environment and seek out open frequencies to communicate. Even in licensed bands, most of the spectrum is empty most of the time. Agile radios could take advantage of that empty space, moving out of the way when another transmission appears.

All of these areas are the subject of extensive academic research and corporate R&D. Nonetheless, the licensed spectrum model has been the dominant paradigm for so long that there is a surprising amount we simply don’t know about how radios work. For example, we don’t know as a theoretical matter what the maximum capacity is of a geographically defined system filled with randomly distributed radios.

We do know that many of our intuitions are wrong. Research has shown that many factors we believe should decrease the capacity of a system—adding more transmitters, creating more alternative paths for signals to travel, or putting receivers in motion, for example—can actually increase capacity. This occurs because the more data a smart receiver has about the surrounding environment, the better it can do in distinguishing the desired signal.

The commercial viability of any system using these techniques will depend on business conditions. That is one reason government policies should advance both designated unlicensed bands and underlay approaches that coexist in licensed bands. Under any scenario, though, open spectrum is not a fantasy, but a concept based on proven techniques. It is time for our policies to catch up with the state of technology.

The Myth Of Scarcity

To take advantage of the fantastic potential of open spectrum, we